

Two experimental studies of children's comprehension of presuppositional *you* 'again' in Mandarin*

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Abstract

Previous studies of children's comprehension of the presupposition of English 'again' and its Mandarin equivalent 'you' (requiring that an event of the same kind previously took place) have reported successful acquisition by preschool age, using Truth Value Judgment Tasks (TVJTs). However, presupposition failure does not elicit a clear 'false' response from adults, making the TVJT less than optimal for assessing children's knowledge of presuppositions. This aspect of the methodology could explain why some of the previous studies excluded a large proportion of children for seemingly ignoring the presupposition trigger. In this study, we examine Mandarin-acquiring children's comprehension of the presupposition of *you* 'again', using two alternative tasks that target the presupposition without requiring participants to judge the truth or falsity of sentences involving presupposition failure. In the Question-Answer task (Experiment 1), participants listened to stories and answered a *wh*-question after each story. The results revealed that most children responded as though they ignored the presupposition trigger *you*, which highlights the methodological challenge of investigating children's knowledge of presupposition. We then conducted a Felicity Judgment task (Experiment 2), which highlighted the presence of the trigger by asking participants to decide which of two speakers—one who produced *you* and one who did not—answered better. In this task, children displayed sensitivity to the presupposition of *you*, albeit not at adult-like levels. We discuss the implications of these findings in relation to children's knowledge of the obligatory presence of *you* when its presupposition is satisfied, a potential ambiguity in its interpretation, and potential differences between children and adults in how they project presuppositions from *wh*-questions.

Keywords: presupposition, again, preschoolers, Question-Answer task, Felicity Judgment task

1. Introduction

This study concerns children's acquisition of presupposition, a kind of backgrounded information that a speaker takes for granted when making the utterance (Stalnaker 1973), as opposed to information that is explicitly asserted. The question of how children recognize and process this type of linguistic inference has received increasing attention in the field of language acquisition. Our focus is on the repetition-denoting adverb 'again', which triggers the presupposition that a

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prior event of the same kind has previously occurred, as illustrated by the English example (1) and its Mandarin translational equivalent (2).

(1) Mary sneezed again.

(2) Mali **you** da-le penti.¹
Mary again hit-ASP sneeze
‘Mary sneezed again.’

Presupposition of (1) and (2): Mary sneezed at an earlier time.

We can test the status of the presupposition by using standard presupposition projection tests (see Langendoen & Savin 1971). For example, the presupposition that Mary sneezed before is preserved in polar questions, as shown in the English example (3) and its Mandarin counterpart (4) (where *ma* is a sentence-final particle used in polar questions).

(3) Did Mary sneeze again?

(4) Mali **you** da-le penti ma?
Mary again hit-ASP sneeze SFP
‘Did Mary sneeze again?’

Presupposition of (3) and (4): Mary sneezed at an earlier time.

‘Again’ has accordingly been characterized as a presupposition trigger. According to some analyses (e.g., Kamp & Rossdeutscher 1994; von Stechow 1996), the adverb is truth-conditionally vacuous, contributing only to the presupposition of the sentence in which it appears. In other words, a sentence with ‘again’ has the same asserted content as the equivalent sentence without it, but additionally presupposes a prior occurrence of the modified event. Under such an analysis, (1) and (2) assert that Mary sneezed, and presuppose that Mary sneezed at an earlier time.

Previous comprehension studies on children’s acquisition of this presupposition trigger report successful acquisition by preschool age (for English, see Xu & Snyder 2017; for Mandarin, see Liu 2009, 2015; Liu, Pan & Hu 2011; Xu, Snyder & Christie 2022, 2024). However, closer examination reveals that this conclusion is typically based on performance in a Truth Value Judgment Task (TVJT) (Crain & Thornton 1998), in which children listened to short stories and evaluated a puppet’s utterances containing ‘again’ as either true or false. However, the presuppositional nature of ‘again’ presents a methodological challenge for the TVJT, as presupposition failure does not always elicit a clear ‘false’ response from adults, rather sometimes evoking a “squeamish” feeling – neither true nor false (Strawson 1950, 1964; von Fintel 2004; among others). For example, in a context in which Mary did not previously sneeze, it is intuitively difficult to assess what the truth value of the sentences (1) and (2) should be. Some theories characterize this as a truth value gap (e.g., Strawson 1950, 1964; Heim & Kratzer 1998; von Fintel 2004). Given the TVJT is meant to elicit truth value judgments (true vs. false), the TVJT may thus be less suitable for assessing one’s knowledge of presupposition.

In the current study, we employ two alternative tasks to examine Mandarin-acquiring children’s comprehension of the presupposition of *you* (2), one of the translational equivalents of

¹ Throughout the paper, we use boldface to indicate stress.

English *again*.^{2,3} The first experiment employed a Question-Answer task, in which participants listened to short stories and answered a *wh*-question after each story. To anticipate, while this task avoids the problem of asking participants to judge the truth or falsity of a sentence with a presupposition failure, the results run contrary to those of previous studies: preschool-aged children do not appear to interpret Mandarin *you* in an adult-like manner when embedded in a *wh*-question. To address some potential reasons for this finding, we next conducted a Felicity Judgment task, which highlighted the presence of *you* by asking participants to decide which of two speakers—one who produced *you* and one who did not—provided a better answer. To preview, the results from this task indicate that preschool-aged children successfully understand the presuppositional trigger (albeit not at fully adult-like levels). Overall, the findings from the two experiments suggest that preschool-aged children possess nuanced knowledge of the presupposition of *you* ‘again’, and moreover that the non-adultlike behavior observed in the Question-Answer task should not be attributed to a lack of knowledge.

The rest of the paper is organized as follows. Section 2 provides a brief introduction to the relevant background. Sections 3 and 4 present the two experiments. Section 5 discusses the implications of the findings, as well as directions for future research.

2. Previous studies on the acquisition of ‘again’

Developmental studies on the acquisition of ‘again’ are relatively limited. Spontaneous production data reveal that children begin using ‘again’ felicitously to denote repetition before the age of 3;06 (Penner, Tracy & Weissenborn 2000; Lee 2005; Xu & Snyder 2017; Xu, Snyder & Christie 2024). However, these results are based on rather small sample sizes: three Cantonese-acquiring children in the study by Lee (2005), four English-acquiring children in the study by Xu & Snyder (2017), and two Mandarin-acquiring children in the study by Xu, Snyder & Christie (2024).⁴

Liu (2009, 2015) and Liu, Pan & Hu (2011) present a series of experiments investigating children’s knowledge of repetition-denoting *you*, using both production and comprehension measures. In their elicited production task, 17 older children (age range: 2;06-3;10) and 5 younger children (age range: 2;00-2;06) were presented with contexts in which an event (e.g., a boy tossing a certain ball; something falling onto a table) occurred twice.⁵ Each time an event took place, they were asked what happened. The authors report that 2- and 3-year-old children were able to correctly produce *you* when they saw an event happen for the second time.

To assess children’s comprehension of *you*, the researchers used a TVJT for children aged 3;06

² Apart from *you*, Mandarin *zai* can also denote repetition, though it has a different distribution (see Lü 1999, Lin & Liu 2009; Liu & Yip 2023 for discussion). We restrict the scope of our study to *you*, leaving *zai* for future investigation.

³ In addition to denoting repetition, Mandarin *you* has other, less frequent interpretations, such as temporal continuation, addition, and rhetorical uses (see Lü 1999). There is some controversy regarding the core meaning of *you* and how many distinct readings of *you* there are (see Shao & Rao 1985; Biq 1988; Shi 1990; Lü 1999), but the repetition-denoting meaning is the most typical and salient interpretation, particularly when *you* is stressed. We thus set aside these other readings of *you* in the current paper, as they fall outside the scope of the study.

⁴ Penner, Tracy & Weissenborn (2000) focus on the acquisition of the German particle *auth* ‘also’, observing that it appears quite early in development, already at the two-word utterance stage. While they focus on *auth*, they also note that *wieder* ‘again’ is structurally similar and appears early in development.

⁵ Materials were slightly different for children above and below 2;06: the older group watched pre-recorded videos, while the younger group watched the experimenters act out the events with toys.

to 3;11 (n=9). Participants listened to short stories in which an event occurred only once (e.g., a panda ate a banana), and then were asked to judge a puppet's statement containing *you* (e.g., *xionghao you chi-le xiangjiao* 'The panda ate a banana again'). Like adults, most children rejected such sentences, suggesting that they understood the repetition-based meaning of *you*. For younger children (aged 2;00-3;06, n=14, with 7 two-year-olds and 7 three-year-olds), the task was made more interactive. The child would perform an action, e.g., putting a toy car in a box. The investigator would then ask the child, "What did you put in the box?" After the child answered, the investigator would ask, "Did you put a car in the box again?" (using the particle *you*). If the child said 'no' or carried out the action a second time, the investigators concluded that the child understood the meaning of stressed *you*. The authors report that most participants answered 'no', leading them to conclude that children possess knowledge of repetition-denoting *you* from an early age.

However, as previously mentioned, asking for a binary 'true'/'false' or 'yes'/'no' response (whether in the context of the TVJT or in answering the investigator's polar question) may not be ideal for testing children's comprehension of presuppositions. Since presupposition failures do not clearly correspond to a 'false' response, it is unclear how children should react in such truth value gap scenarios. The issue becomes particularly acute in the studies by Xu (2015), Xu & Snyder (2017) and Xu, Snyder & Christie (2022, 2024), which examine English- and Mandarin-acquiring children's knowledge of 'again'/*you*. These studies were focused on the so-called restitutive interpretation of 'again' (e.g., von Stechow 1995, 1996; Beck & Johnson 2004; Beck 2005), which is not relevant for our study, but they did include a control condition that tested the repetition-denoting meaning. In their TVJTs, participants listened to a series of short stories, each involving two characters, only one of whom satisfied the presupposition of 'again'. For example, in one story, a dinosaur crawled under a tree twice, while a crocodile did so only once. After hearing the story, participants were asked to judge a puppet's utterance, which was either about the character who had repeated the action, making the use of 'again' felicitous ('the dinosaur crawled under the tree again'), or about the character who only performed the action once, making the use of 'again' infelicitous ('the crocodile crawled under the tree again'). Preschool-aged children successfully accepted the felicitous sentences and rejected the infelicitous ones, suggesting that they correctly understood the repetitive reading of 'again'. However, this finding was based on a subset of the collected data, as up to 40% of child participants were excluded for seemingly ignoring 'again' – that is, they accepted (almost) all of the test sentences containing 'again'. The authors speculated that because the assertion of these test sentences matched the given context, children might have downgraded the relevance of presupposition and prioritized the true assertion. It is also possible that children noticed the presupposition failure, but were more pragmatically tolerant (see Katsos & Smith 2010; Davies & Katsos 2010) and thus accepted the test sentences, even though they recognized that they were not perfect descriptions. Ultimately, the authors acknowledge that the TVJT might not be the most effective method for accessing children's knowledge of presupposition.

Our study aims to investigate Mandarin-acquiring children's comprehension of the presupposition of *you* (2), using alternative tasks that, importantly, avoid asking participants to judge the truth or falsity of a sentence that involves presupposition failure. We turn now to Experiment 1, which involved a Question-Answer task.

3. Experiment 1: Question-Answer Task

3.1. Methods

3.1.1. Participants

40 Mandarin-acquiring children (including one 3-year-old, 23 4 year-olds and 16 5-year-olds ranging from 3;11 to 5;12, mean age: 4;10, $SD = 0.52$, 20 girls) and 34 adult native speakers of Mandarin participated in the experiment. An additional 11 children were tested but excluded because they failed to meet our inclusion criterion of scoring at least 75% accuracy on fillers, and an additional five children were excluded because a parent interfered with the session.

Both adult and child participants were recruited and tested individually online (via Tencent Meeting) during the COVID-19 pandemic.

3.1.2. Procedure

Participants listened to a series of pre-recorded, illustrated short stories, each featuring two contrasting characters engaging in some activity before and after the sun set. We reasoned that sunset was a time point for which it was relatively easy to visually depict a ‘before’ and ‘after’ period; moreover, children should be familiar with the concept of sunset (as opposed to, for example, a particular time or a particular day of the week).

Between the two characters in each story, only one satisfied the presupposition of *you*, by carrying out an action twice. At the end of each story, participants were presented with a pre-recorded *wh*-question, which asked who had performed the relevant action. Participants had to choose between the character who had done the action once, the character who had done the action twice, and a ‘both’ option if they felt that both characters had done the relevant action. We provide a translated example of a target trial in (5) (paired with Figure 1), but keep the critical *wh*-question in Mandarin. In the *wh*-questions on the target trials, *you* was stressed in a natural way to make the adverb more salient.

- (5) *Story*: Today Cat and Fox were playing with some very ticklish magical feathers in the forest. They made up a game: even if a feather tickles you, you can’t sneeze! Cat tickled Fox’s nose with a feather. Fox let out a big sneeze, “Ahhh-choo!” Then Fox tickled Cat’s nose with a feather. Cat managed not to sneeze. Soon, the sun set and it was evening. The two friends decided to play one more round. Cat found a feather and tickled Fox’s nose. Fox let out a big sneeze, “Ahhh-choo!” Then Fox tickled Cat’s nose with a feather. Cat let out a big sneeze, “Ahhh-choo!”

Experimenter: We know Cat and Fox played with feathers before and after the sun set.

Taiyang	luo-shan	hou	shui	you	da-le-penti:
Sun	fall-mountain	after	who	again	hit-ASP-sneeze
huli,	xiao-mao,	haishi	taimen	liang-ge?	
fox	little-cat	or	they	two-CL	

‘Who sneezed again after the sun set: Fox, Cat, or both?’

Target response: Fox



Figure 1: Pictures paired with example (5), depicting the outcome of the sneezing game before (left) and after (right) the sun set.

3.1.3. Materials

We tested four different predicates (*da-penti* ‘sneeze’, *shuai-dao* ‘fall’, *shui-zhao* ‘fall asleep’, *pao-bu* ‘run’). The target trial with the predicate ‘sneeze’ is presented in (5). If a participant was sensitive to the presupposition of *you*, they were expected to choose the character who performed the action twice. The expected answer for (5) was thus Fox, as he had sneezed earlier, before the sun set. Conversely, if participants were not sensitive to the presupposition of *you* or completely ignored it, they were expected to select the ‘both characters’ option.

For each of the predicates that appeared in the target condition, we also included a control trial in which the *wh*-question did not contain *you* ‘again’ (see example (6), paired with Figure 2). Participants were asked “Who sneezed after the sun set?” In this case (in the absence of *you* ‘again’), the expected answer was ‘both’ characters.

- (6) *Story*: Today Giraffe and Duck were tidying the kitchen, which was very dusty. They each moved a box from the corner to the middle of the kitchen. Duck accidentally dropped his box on the floor, and there was a big cloud of dust. Duck let out a big sneeze, “Ahhh-choo!” Giraffe gently set his box down and managed not to sneeze. Soon, the sun set and it was evening. The friends decided to move the last boxes. Duck tried to set his box down gently, but still there was a cloud of dust. Duck let out a big sneeze, “Ahhh-choo!” Then Giraffe accidentally dropped his box on the floor, and there was a big cloud of dust. Giraffe let out a big sneeze, “Ahhh-choo!”

Experimenter: We know Duck and Giraffe moved some boxes before and after the sun set.

Taiyang luo-shan hou shui da-le-penti:
Sun fall-mountain after who hit-ASP-sneeze

Xiao-ya, changjinglu, haishi taimen liang-ge?
little-duck giraffe or they two-CL

‘Who sneezed after the sun set: Duck, Giraffe or both?’

Target response: Both

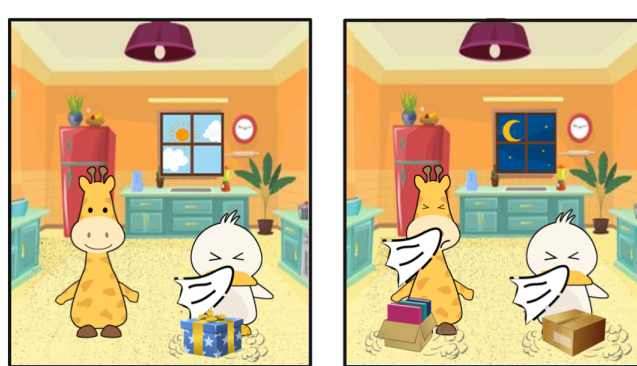


Figure 2: Pictures paired with example (6), depicting the outcome of moving boxes before (left) and after (right) the sun set.

All participants received two training items in the same order, followed by 16 trials (four critical *you* targets, four controls without *you*, and eight fillers) in a blocked design, with the targets preceding the controls. This blocked order was chosen to ensure that the absence of *you* on control trials would not influence participants' interpretation of the target trials. Without a block design, the critical targets and controls would appear in a randomized order. If a participant encountered a control trial without *you* first, this could encourage them to (continue to) respond in a manner consistent with the absence of *you*. Four filler items were interspersed within each block, leading to a total of 8 trials in each block.

The training and filler items were simple stories, each involving two characters engaging in activities, followed by a *wh*-question at the end asking who had done something. For instance, in the story where Chicken bought a guitar and a violin, and Sheep bought a piano and a violin, participants were asked “Who bought a guitar – Chicken, Sheep or both?” Among the eight filler items, two had ‘both’ as the target answer, and the other six had one of the characters as the target answer. This ensured that the three possible answers (the first character in the *wh*-question, the second character in the *wh*-question, and ‘both’ characters) were roughly equal in number.

We counterbalanced whether the expected target character appeared to the left or the right, to control for potential preferences for a particular side. The trials within the target and control blocks were fully randomized across participants using a Python script.

3.2. Results

Figure 3 displays the proportion of selections of the ‘repeater’ character, i.e., the character who performed the action twice, in the control and target conditions, across children and adults. We can see from the figure that adults, as expected, tended to select the repeater in the target condition, but not in the control condition. Children’s behavior did not mirror that of adults: while they tended not to select the repeater in the control condition, they did not show a tendency to select it in the target condition either.

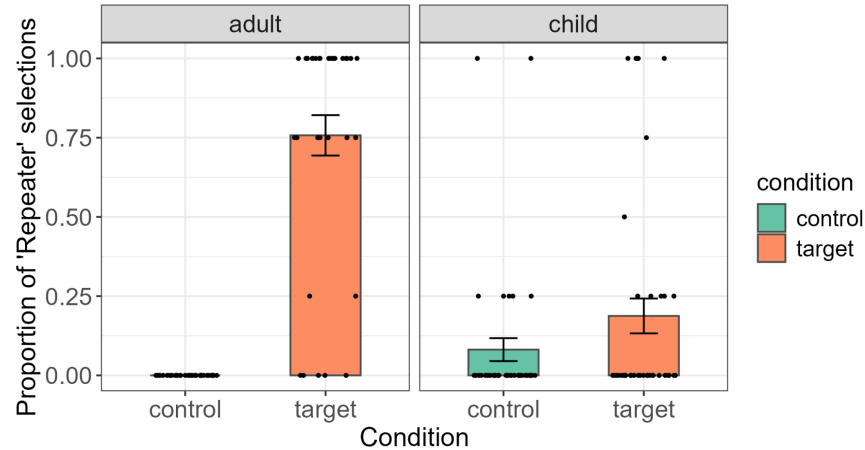


Figure 3: Proportion of selections of the ‘repeater’ character in control and target conditions, across children and adults. Error bars represent the standard error of the mean, and individual dots indicate each participant’s mean proportion of selecting the repeater character.

None of the adult participants selected the repeater character in the control condition. The lack of variance in this condition precluded the use of ANOVAs or linear models for analyzing the full dataset. We thus conducted two Mann-Whitney U tests on the target and control conditions, respectively. The results for the target condition indicate that children selected the repeater character significantly less than adults did ($W = 1122.5, p < .001$, Bonferroni-adjusted for multiple comparisons). For the control condition, the proportion of selecting the repeater differed significantly across the two participant groups ($W = 561, p = .011$, Bonferroni-adjusted for multiple comparisons). The 95% confidence interval for the proportion of children selecting the repeater character in the control condition (0.01 – 0.16) did not cross 0, suggesting the same pattern.

Figure 4 provides a plot of the proportions of each response type given by children and adults in the target and control conditions. The plot shows that when adults and children did not select the repeater character in the target condition, the majority of them opted for the ‘both’ option. This would be the expected answer if the question did not contain *you* ‘again’. In such cases, the question essentially inquires about what happened after the sun set, and in all cases, both characters performed the action after the sun set.

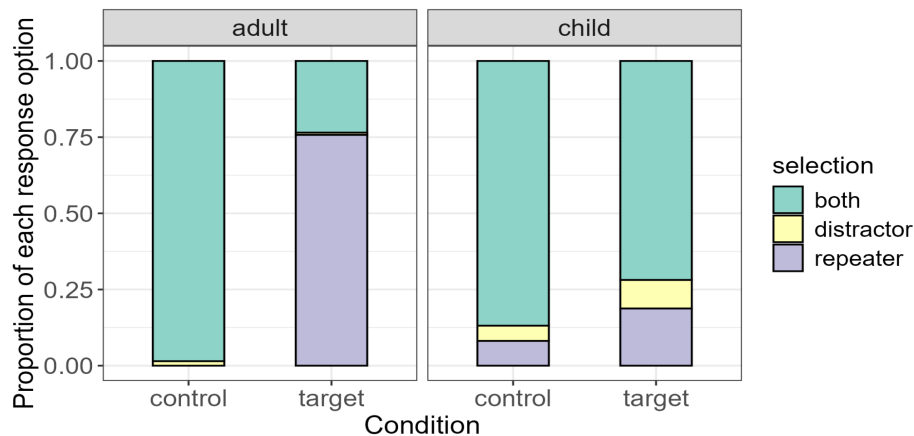


Figure 4: Proportion of selections of each response option.

To better understand children's selections in the target and control conditions, we next analyzed the child data using multinomial logistic regression, using the *mclogit* package (Elff 2022) in R version 4.3.1 (R Core Team 2021).⁶ For the three-level outcome variable (both, distractor, repeater), we set the response ‘both’ as the reference category. We considered models with varying random effect structures, including models with and without random intercepts and random slopes for Subject and Item. For fixed effects selections, we used the best subset regression method (Miller 2002), considering models with all possible combinations of the following predictors: Condition (target vs. control), Age (as a continuous variable in years and centered) and their interaction. Model fit was evaluated based on Akaike Information Criterion (AIC), following Bolker et al. (2009). The model with the lowest AIC included Condition and Age as predictors, and a random intercept for Subject (Selection ~ 1 + Condition + Age + (1 | Subject)). The results, summarized in Table 1, showed that children were more likely to select the repeater on target trials compared to control trials. For the ‘distractor’ responses, children also favored this choice more on target trials. There was no Age effect for either the ‘both’ or ‘distractor’ responses.

Table 1. Results of multinomial logistic regression model. The outcome variable was coded with ‘both’ as the reference level.

	β	<i>SE</i>	<i>z</i>	<i>p</i>	OR [95% CI]
repeater vs. both					
Condition (target)	2.36	0.58	4.10	<.001	10.55 [3.42, 32.56]
Age (centered)	0.21	0.99	0.21	.834	1.23 [0.18, 8.50]
distractor vs. both					
Condition (target)	1.99	0.69	2.89	.004	7.28 [1.89, 28.00]
Age (centered)	-2.06	1.11	-1.85	.064	0.13 [0.02, 1.13]

3.3. Discussion

Experiment 1 used a Question-Answer Task designed to avoid asking participants to judge the truth/falsity of sentences exhibiting presupposition failure. However, the results ran counter to the previous studies: while children exhibited some sensitivity to *you*, their performance was not adult-like. Most of them selected the ‘both’ option, as if ignoring the presupposition trigger.⁷ Even among adults though, we observed some proportion of selecting the ‘both’ option.

Let us consider first why adults might have opted for the ‘both’ option. One possibility may be related to the so-called ‘agentless’ presupposition reading of ‘again’, which requires only that the action be repeated, regardless of who performs it (see Bale 2007; Ausensi, Yu & Smith 2021;

⁶ We did not fit multinomial logistic regression models to the adult or combined data, because no adult participants selected the repeater in the control condition.

⁷ Importantly, this tendency cannot be explained by children quickly deciding on their answer before they have even heard the presupposition trigger, since in Mandarin *you* is preverbal (Jeff Lidz, p.c.).

Smith & Yu 2022, 2024; Zhang 2022; Asami & Bruening 2025). This reading is illustrated in (7), where there are two events of guessing the riddle wrong, but crucially each involving a different agent – first the big brother, and then the little brother.

- (7) Gege cai-cuo-le, didi you cai-cuo-le.
 Big.brother guess-wrong-ASP little.brother again guess-wrong-ASP
 Zhe-ge miyu ke nan le.
 this-CL riddle rather difficult SFP
 ‘The big brother guessed the riddle wrong, and so did the little brother. This riddle is quite difficult.’

(Zhang 2022, 88: (12a))

To illustrate how this reading might have been at play in the experiment, consider the example story in (5), where two sneezing events took place: only Fox sneezed before the sun set, but both Cat and Fox sneezed after the sun set (hence Fox sneezed twice, while Cat sneezed only once). This story was followed by a question about who sneezed again after the sunset. Under the agentless presupposition reading, the *wh*-question presupposes that a prior sneezing event occurred before the sunset, which is clearly satisfied in the story (by Fox’s sneezing before the sunset), and asks about the agent of the sneezing event that took place after the sunset. Some of the adult (and child) participants may have accessed the agentless presupposition reading, leading them to respond “both”. That there were relatively few ‘both’ selections in adults suggests this reading is not very salient, or at least is not the predominant reading; on the other hand, if this reading is indeed responsible for the ‘both’ response, it seems that children were much more willing to respond on the basis of this reading, compared to adults.

Why might children have selected ‘both’ more often than adults? Setting aside the possibility of an agentless presupposition reading, children at this age might simply lack the relevant presuppositional knowledge of *you*, causing them to disregard the trigger.

Another possibility is that the presupposition of *you* was not made sufficiently salient in the experimental contexts. Although *you* was stressed in the *wh*-question, it might not have been prominent enough for child participants to pay attention to it. Even without considering the trigger, they would still have been able to arrive at an interpretation and provide an answer.

Yet another possibility centers on participants’ ability to project the presupposition from the *wh*-question. It has been suggested that presuppositions in *wh*-interrogatives project universally (Schlenker 2008; Abrusán 2014; Nicolae 2015). According to this view, a *wh*-question such as (8) would presuppose that all the individuals in the domain (i.e., Cat and Fox) sneezed before.

- (8) Who sneezed again after the sun set: Cat, Fox, or both?

Presupposition: All individuals in the domain (Cat, Fox) sneezed before.

Meanwhile, certain pragmatically non-canonical questions, such as quiz questions, consistently exhibit projection patterns weaker than universal projection (Schwarz & Simonenko 2018; Theiler 2020). To characterize the contrast between canonical information-seeking questions and non-canonical questions in their presupposition projection patterns, Theiler (2020) proposes what she calls the “Epistemic Bridge”, as described in (9).

- (9) A question *Q* is felicitous relative to a context set *c* only if the presuppositions of all answers that the speaker considers possibly true are entailed by the context set. (Theiler 2020: (22))

The intuition behind (9) is that the speaker of a question wants to avoid a situation in which the desired common ground update cannot take place because a true answer turns out to be infelicitous. For a canonical information-seeking question, because the speaker is agnostic about the possible true answers, universal projection of the presupposition would ensure that all the true answers are also felicitous. In contrast, for non-canonical information-seeking questions, the speaker already has prior knowledge about which questions will turn out to be true. As a result, the question allows for (weaker) existential projection of embedded presuppositions.

In our experiment, the test questions containing *you* are not canonical information-seeking questions. The speaker (the experimenter) already knows which answer is true and felicitous. In other words, one can draw a prior knowledge implicature. Adult participants, as addressees, can make this prior knowledge inference and restrict the domain of answers to those the speakers considers possibly true. Unlike adults, children may not restrict the domain of answers in the same way, instead adhering to a universal-projection interpretation, which leads to presupposition failure (e.g., in (5), it is false that both Cat and Fox sneezed earlier). Consequently, children ignore ‘again’ and select “both”.

In Experiment 2, we moved to a Felicity Judgment Task, which improved upon Experiment 1 by ensuring sufficient salience of the trigger by using direct contrast to highlight the presence of *you*. The second experiment also circumvented the issue of presupposition projection by testing the interpretation of *you* in declarative sentences instead of *wh*-questions, all the while not requiring participants to judge the truth or falsity of a sentence with presupposition failure. Additionally, in order to assess the felicity of *you*, participants had to determine whether a mentioned character had repeated the action or not, making it less likely for them to access the agentless presupposition reading.

4. Experiment 2: Felicity Judgment Task

Experiment 2 builds on the observation that presupposition triggers like the additive particle *too*, the iterative particle *again*, and the definite determiner *the*, are obligatory when their presuppositions are satisfied in the context (see Percus 2006; Chemla 2008; Sauerland 2008 for English, Heim 1991; Bade 2016 for German, Amsili, Ellsiepen & Winterstein 2016 for French). This is illustrated in (10), where the absence of English *again* results in infelicity. A similar pattern is observed for the Mandarin translational equivalent *you*, as shown in (11).

- (10) Jenna went ice skating yesterday. She went ice skating today, #(again).
(Bade & Renans 2021: (1b))

- (11) Zhangsan zuotian qu-huabing le. Jintian #(you) qu-huabing le.
Zhangsan yesterday go-skate ASP today again go-skate ASP
‘Zhangsan went skating yesterday. He went skating today, #(again).’

We will return to examples like (10) and (11) in the Discussion, after we have presented Experiment 2.

4.1. Methods

4.1.1. Participants

43 Mandarin-acquiring preschool-aged children (30 4-year olds and 13 5-year-olds, age range: 4;01–5;12, $M=4;10$, $SD = 0.495$, 18 girls) and 33 adult native speakers of Mandarin participated in the experiment.⁸ An additional six children were excluded because they failed to meet our inclusion criterion of scoring at least 75% accuracy on fillers.

4.1.2. Procedure

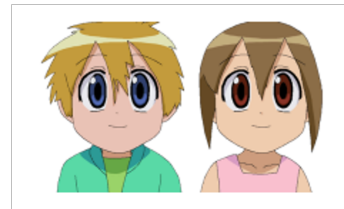
We implemented a Felicity Judgment Task in which participants watched the same set of stories as those in Experiment 1. After each story, the experimenter asked a question and two speakers (a boy and a girl) answered the question. Participants were asked to determine who had answered better; see example (12).

(12) *Story*: [same as in (5)]

Experimenter: We know Cat and Fox played with feathers before and after the sun set. What happened after the sun set?

Boy: Huli **you** da-le-penti.
Fox again hit-ASP-sneeze
'Fox sneezed again.'

Girl: Huli da-le-penti.
Fox hit-ASP-sneeze
'Fox sneezed.'



Experimenter: Who answered better?

Target response: Boy

The experiment was implemented using Qualtrics. The stories and the two speakers' responses were pre-recorded. The adult participants completed the experiment independently via a provided link, where they could read the question after each story. The child participants were tested individually by the experimenter, who read the question aloud after each story.

4.1.3. Materials

We had both a target and a control condition in the experiment; in both conditions, we tested the same four predicates as in Experiment 1: *da-penti* 'sneeze', *shuai-dao* 'fall', *shui-zhao* 'fall asleep', and *pao-bu* 'run'.

As illustrated in (12), for the target trials, the boy and the girl each answered the experimenter's question by mentioning the character who had repeated the action. One speaker used *you* 'again' while the other did not. The expected response on these trials was the speaker who had used the trigger *you* 'again', because its presupposition was satisfied, thus making its presence obligatory.

In the control condition, both speakers answered the experimenter's question by mentioning the character who had performed the action only once. The expected response was to select the

⁸ None of the participants had completed Experiment 1.

speaker who had not used *you* ‘again’, because the utterance containing *you* would involve a presupposition failure; see example (13).

(13) *Story*: [same as in (6)]

Experimenter: We know Duck and Giraffe moved some boxes before and after the sun set. What happened after the sun set?

Boy: Changjinglu **you** da-le-penti.
 Giraffe again hit-ASP-sneeze
 ‘Giraffe sneezed again.’

Girl: Changjinglu da-le-penti.
 Giraffe hit-ASP-sneeze
 ‘Giraffe sneezed.’

Experimenter: Who answered better?

Target response: Girl

All participants received two training items, followed by 16 trials: four targets, four controls, and eight fillers, presented in a randomized order. We counterbalanced which speaker corresponded to the target response and which speaker responded first, to control for any potential preferences children might have for one of the two speakers.

In the filler items, the two speakers also answered a question by mentioning a character, with their responses differing in truth value. For example, in a story in which Chicken bought a guitar and a violin, and Sheep bought a piano and a violin, the experimenter asked, “We know Chicken and Sheep each bought a violin. What else did they buy? The boy responded, “Sheep bought a piano” (true) while the girl responded, “Sheep bought a guitar” (false). Participants were then asked to choose who had answered better.

4.2. Results

Figure 5 displays the proportion of responses selecting the speaker who used *you* ‘again’, across control and target conditions, in children and adults.

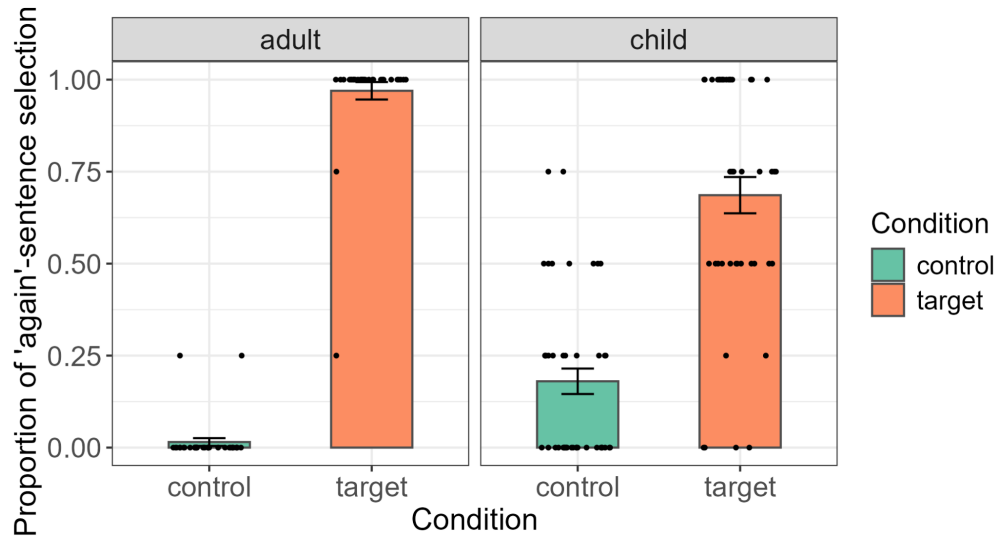


Figure 5: Proportion of responses selecting the speaker who uttered *you* ‘again’ in control and target conditions, across adult and child participants. Error bars represent the standard error of the mean, and individual dots indicate each participant’s mean proportion of selecting the speaker who used *you*.

We analyzed the data with a mixed effect logistic regression model with the lme4 package (Bates et al., 2015) in R. The dependent variable was children’s responses (1=selecting the again-speaker, 0=not selecting the again-speaker). Fixed effects included Condition (target vs. control), ParticipantGroup (adult vs. child), and their interaction. Following Barr et al. (2013), we started with a model with a maximal random effect structure (including random slopes and intercepts for Subject and Item), and then reduced the model’s complexity stepwise until it converged to a non-singular fit. This resulted in a model that included by-participant random slopes and intercepts. Likelihood ratio tests indicated that adding the interaction significantly improved model fit ($\chi^2(1) = 34.49, p < .001$). We therefore included all three fixed-effect variables in the final model: $\text{Selection} \sim 1 + \text{Condition} * \text{ParticipantGroup} + (1 + \text{Condition} | \text{Subject})$. Model results revealed a significant effect of Condition ($\beta = 9.71, SE = 1.29, z = 7.55, p < .001$) and ParticipantGroup ($\beta = 2.83, SE = 0.80, z = 3.54, p < .001$). In addition, there was a significant interaction between Condition and ParticipantGroup ($\beta = -6.62, SE = 1.30, z = -5.10, p < .001$). These results indicate that both groups significantly favored the speaker who used *you* in target trials compared to control trials, with adults exhibiting a larger difference between targets and controls than children.

Figure 6 displays the different response types given by the different participant groups. The category ‘other’ includes trials where participants either could not give a response or responded by choosing both speakers. Both groups favored the speaker who used *you* in the target condition and the speaker who did not use *you* in the control condition. While children’s responses were noisier than those of adults, their preferred responses mirrored those of adults.

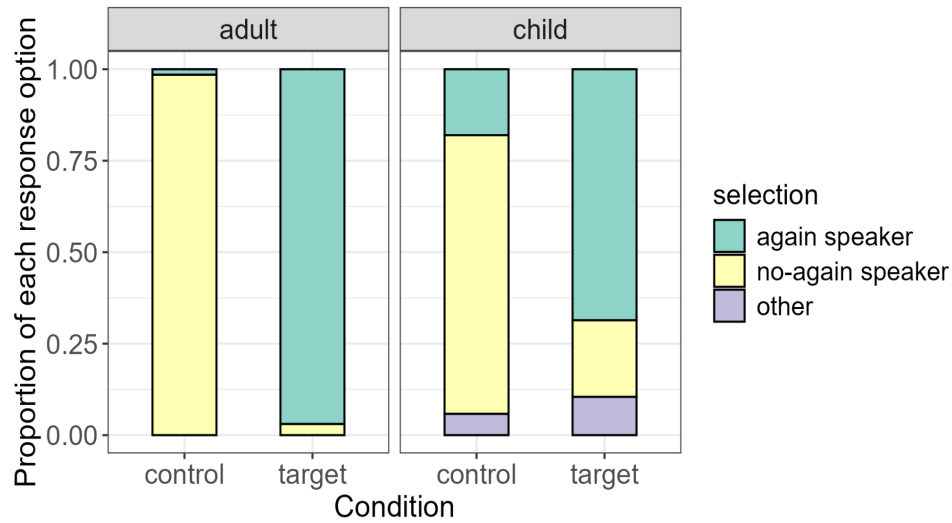


Figure 6: Proportion of selections of each response option. ‘Other’ responses corresponded to trials where participants could not give a response, or answered by choosing both speakers.

To better understand children's selections in the target and control conditions, we next analyzed the child data using multinomial logistic regression. For the three-level outcome variable (selecting the speaker with *you*, selecting the speaker without *you*, and ‘other’), we set the response of selecting the speaker with *you* as the reference level. We considered models with varying random effect structures, including models with and without random intercepts and slopes for Subject and Item. We also considered fixed effect models with all possible combinations of the following predictors: Age (as a continuous variable and centered), Condition (target vs. control), and their interaction. The model with the lowest AIC included all predictors and a random intercept of Subject (Selection ~ 1 + Condition * Age + (1 | Subject)). The model results, summarized in Table 2, suggest that higher age increased the odds of choosing the speaker who did not produce *you* (generally related to improved performance on the controls). Children were overall more likely to choose the speaker who used *you* in the target trials compared to control trials. Moreover, this effect was more pronounced amongst older children. For the ‘other’ responses, on the other hand, there were no significant effects of condition, age, or their interaction.

Table 2. Results of the multinomial logistic regression model. The outcome variable was coded with the ‘*again-speaker*’ as the reference level.

	β	SE	z	p	OR [95% CI]
no-again-speaker vs. again-speaker					
Condition (target)	-3.04	0.32	-9.41	<.001	0.05 [0.03, 0.09]
Age (centered)	1.85	0.66	2.80	.005	6.36 [1.74, 23.17]
Age: Condition	-2.63	0.73	-3.62	<.001	0.07 [0.02, 0.30]

other vs. <i>again</i> -speaker					
Condition (target)	-0.57	0.52	-1.11	.265	0.56 [0.21, 1.55]
Age (centered)	1.47	1.11	1.32	0.187	4.34 [0.49, 38.42]
Age: Condition	-2.01	1.20	-1.68	0.093	0.13 [0.01, 1.40]

4.3. Interim summary

Experiment 2 shows that children successfully understand the presuppositional trigger *you* ‘again’, though not yet at fully adult-like levels. As in Experiment 1, the task avoids the problem of forcing participants to judge the truth or falsity of sentences exhibiting presupposition failure, and unlike previous studies, we did not have to exclude children who appeared to ignore the trigger. Furthermore, this method set aside some of the issues we identified as potentially leading to children’s selection of the ‘both’ response in Experiment 1. First, it resolved the issue of insufficient saliency of the trigger: the only difference between the two speakers’ descriptions was the presence/absence of *you*. If participants ignored the trigger, they would be comparing two identical sentences. Second, since both speakers focused on one animal character when responding to the experimenter’s questions, participants needed to consider whether the mentioned character had repeated the action in order to assess whether the use of *you* was felicitous. This setup made it less likely for participants to access the agentless presupposition reading (according to which it does not matter who repeated the action). Moreover, since participants were asked to compare two declarative sentences, the task overcame the issue of children’s potential difficulty with presupposition projection from *wh*-questions.

5. General Discussion

In this paper, we conducted two experiments to tap into Mandarin-speaking children’s knowledge of the presupposition trigger *you* ‘again’. The results of the Question-Answer Task suggest children were simply ignoring the presupposition trigger, while the results of the Felicity Judgment Task suggest sensitivity to the presupposition (albeit not at an adult-like level). In this section, we discuss the broader implications of our findings and highlight some questions for future research. First, we discuss the results of the Felicity Judgment Task in the context of children’s knowledge of the obligatory presence of a trigger when its presupposition is satisfied. We then explore in greater depth two underlying factors that may have contributed to children’s selection of the ‘both’ option in the Question-Answer Task: their access to the agentless presupposition reading and their ability to project presuppositions from *wh*-questions.

5.1. On children’s knowledge of the obligatory presence of *you* ‘again’

The results of the Felicity Judgment Task demonstrate that many preschool children are sensitive to the obligatory presence of *you* when its presupposition is satisfied. In the target contexts, in which the relevant character carried out the action twice, children recognized that it was better to describe the situation using the presupposition trigger *you* ‘again’ than without it. Here we consider what knowledge might underlie this observed sensitivity.

Several analyses have been proposed to account for the apparent obligatory presence of the trigger, and here we focus on two major formal accounts (see Aravind & Hackl 2017 for another possibility). The first is *Maximize Presupposition!* (MP), a principle that requires speakers to choose the presuppositionally stronger statement among a set of contextually equivalent expressions (Heim 1991; see also Percus 2006; Chemla 2008; Sauerland 2008; Amsili & Beyssade 2010; Singh 2011; Schlenker 2012). Initially proposed to account for the contrast between the definite determiner *the* and the indefinite determiner *a*, it was subsequently applied to other triggers. For the obligatory presence of ‘again’, the competition lies between a sentence with ‘again’ and one without it. In a context in which the presupposition of ‘again’ is satisfied, *Maximize Presupposition!* leads the speaker to utter the sentence containing the trigger.

An alternative account is Obligatory Inference (OI, see Bade 2016; Bade & Tiemann 2016; Bade & Renans 2021), which argues that the insertion of a trigger is necessary to avoid semantic conflict. Consider (14). In the absence of the trigger, “after the sun set” carries focus, triggering an exhaustive interpretation (i.e., Fox sneezed after the sun set *and at no other time*), which contradicts the preceding sentence. Inserting ‘again’ prevents this contradiction by blocking the exhaustive interpretation, because the presupposition of ‘again’ ensures that a prior event in which Fox sneezed occurred earlier, thus avoiding the conflict.

(14) Fox sneezed before the sun set. He sneezed #(again) [after the sun set]_F.

The results of our experiment are in principle compatible with both theoretical accounts (MP and OI), and suggest that knowledge of the relevant mechanism is already emerging in 4- and 5-year-old children. Our experimental study was not designed to tease apart the two theoretical accounts. The findings, however, do raise questions about how child learners might develop the observed sensitivity to the obligatory presence of a presupposition trigger.

Developmental research on this issue has primarily focused on MP, examining different triggers and yielding mixed findings. Some studies suggest early mastery of MP (see, e.g., Wexler 2003 on children’s acquisition of the definite determiner *the* and indefinite determiner *a*), while others point to later mastery (e.g., Yatsushiro 2008 on German-speaking children’s comprehension of the universal quantifier *jeder* ‘every’ and the definite determiner *der/die/das* ‘the’), particularly in comparison with lexically encoded presuppositions (e.g., Legendre et al. 2011 on 30-month-olds’ acquisition of French personal pronouns). Recent investigations further complicate the picture, showing mixed performance for different triggers (see Aravind 2018 on preschoolers’ acquisition of *both* and *another*, and Torma, Brody & Aravind 2021 on preschoolers’ comprehension of *both* and *either*).

Because our results are compatible with both MP and OI, they do not directly determine whether MP is in place early or late. However, if MP indeed governs the obligatory presence of ‘again’, our results suggest that the non-adultlike performance observed in previous studies of other MP-governed triggers stems not from a lack of knowledge of MP, but rather other factors, such as a lack of knowledge about what forms should compete, or a non-adultlike semantic representation of the triggers themselves. On the other hand, no prior developmental work to our knowledge has examined children’s knowledge of OI, making it an even less explored area than MP. If instead of MP it is OI that governs the obligatory presence of *you* ‘again’, our results could suggest that OI is in place relatively early. Future work could aim to tease apart the two accounts by identifying cases where the two mechanisms make different predictions.

5.2. On children's knowledge of the agentless presupposition interpretation

We proposed earlier that some child (and adult) participants may have selected the ‘both’ option in the Question-Answer task because they accessed the agentless presupposition reading of ‘again’, on which the relevant action is indeed repeated, but can be performed by a different agent. We speculate that the narrative progression of the stories in our experimental items, which involved two distinct time periods—one before and one after the sun set—may have implicitly encouraged some participants to consider two events and focus on the shift in agents across them, making the agentless presupposition reading more salient.

Theoretical analyses argue that the agentless presupposition reading is derived structurally, with repetition-denoting ‘again’ scoping over the predicate while excluding the subject at LF (see Bale 2007; Ausensi, Yu & Smith 2021; Smith & Yu 2022, 2024; Zhang 2022; Asami & Bruening 2025). This is illustrated by the Mandarin example (15), with the derivation of the agentless presupposition reading presented in (15c).

- (15) a. Lili you qie-le na-zhang da-bing.
 Lili again cut-ASP that-CL big-pancake
 ‘Lili cut the big-pancake again.’ (ambiguous)
- b. repetitive reading: Lili had cut the big pancake before.
 [VoiceP Lili [vP again [vP cut that big pancake]]]
- c. agentless presupposition reading: Someone (else) had cut the big pancake before.
 [VoiceP again [VoiceP Lili [vP cut that big pancake]]]
- (adapted from Zhang 2022, 94: (22))

Under such a structural analysis of the agentless presupposition reading, children who access this reading must already have a basic understanding of the repetitive meaning of ‘again’, which aligns with the results of the Felicity Judgment Task. To our knowledge, no prior study has investigated children’s knowledge of this reading, making it a topic for future research.

5.3. On children's ability to project presuppositions from *wh*-questions

Another possible explanation for children's selection of the 'both' response in the Question-Answer Task involves the way they project presuppositions from *wh*-questions. Unlike adults, children may not restrict the domain of possible answers to those the speaker considers possibly true. Instead, they may access a universal projection interpretation, which leads to presupposition failure, leading them to ignore the trigger in the *wh*-questions.

Similar ideas have been proposed regarding the way children interpret presuppositions under quantifiers like *none*. Zehr et al. (2016) examined how adults and children interpret sentences like *None of the bears won the race*, which contains the presupposition trigger *win* (presupposing that the relevant characters *participated* in the race). Their findings indicate that while adults access both the universal projection interpretation (i.e., *every bear participated in the race*) and the existential projection interpretation (i.e., *at least one of the bears participated in the race*), children only access the universal projection interpretation. To reconcile theories of presupposition projection with the experimental data from both adults and children, the authors propose that for presupposition projection from the scope of negative quantificational expressions, universal projection is obtained directly via projection, while existential projection builds upon it and

requires additional mechanisms, such as domain restriction, which the child participants may not yet have mastered. Our experimental findings corroborate this proposal.

If our explanation is on the right track, we predict that children might also ignore other presupposition triggers (such as ‘also’) in similar tasks, where universal presupposition projection in *wh*-questions leads to presupposition failure. Kurokami et al. (2021, 2022) investigated preschool-aged children’s comprehension of English *also/too* and its Japanese counterpart *mo* in *wh*-questions. Participants first listened to stories, in which Mickey, Minnie, and Donald engaged in some activities. In each story, one character completed only one task, another character completed a different task, and a third character completed both. For example, in one story, Minnie ate only an apple, Donald ate only a banana, and Mickey ate both an apple and a banana. At the end of the story, an on-screen puppet was asked who ate the most fruit. The puppet first attempted to recall the story, answering, “Donald didn’t eat an apple, but Mickey and Minnie did.” Then, being forgetful, the puppet asked an incidental *wh*-question to solicit help from the participant, such as “Who (also) ate a BANANA?”. The results showed that children were more likely to select the character who performed both actions in the ‘also’-condition than in the ‘also’-less condition.⁹ Crucially, the proportion of adult-like responses, i.e., child responses selecting the two-action character in the critical trials (41% in the Japanese experiment, 52%-70% in the English experiments) was higher than the proportion of adult-like performance in our Question-Answer Task, which yielded only 25% child responses selecting the repeater in the critical ‘again’-trials.

A closer examination reveals two key differences between Kurokami et al.’s experiments and ours. First, their *wh*-question was not a non-canonical information-seeking question. Rather, the forgetful puppet genuinely sought information from participants. Second, the *wh*-question was embedded within a larger discourse, where the puppet’s partial recall of the story (Mickey and Minnie ate apples) served as a linguistic antecedent that satisfied the presupposition of ‘also’ in the subsequent question ‘Who also ate a BANANA?’ This prior discourse might have restricted the domain of potential answers to the two characters who satisfied the presupposition of ‘also’. Even if children interpreted the presupposition as projecting universally, there was no presupposition failure. As a result, Kurokami et al.’s experimental findings are not inconsistent with our predictions. Future research could more closely examine whether children differ from adults in how they project presuppositions from *wh*-questions.

6. Conclusion

In this study, we implemented two alternative tasks to the TVJT—a Question-Answer Task (Experiment 1) and a Felicity Judgment Task (Experiment 2)—which were both designed to tap into preschool-aged children’s knowledge of the presupposition of Mandarin *you* ‘again’. Both tasks avoided the problem of forcing participants to judge the truth or falsity of sentences exhibiting presupposition failure. Importantly, the results hold without having to exclude children who might appear to ignore the trigger. In contrast to previous findings, the Question-Answer Task showed children responding as if they were ignoring the trigger. We discussed several factors which might have led to their non-adultlike behavior. A follow-up Felicity Judgment Task, which was designed to address some of the issues with the first experiment, revealed that children

⁹ Notably, while Kurokami et al. (2021) reported relatively low proportions of child responses selecting the two-action character in their critical ‘also’-condition (41% in the Japanese experiment, 52% in the English experiment), their follow-up study with methodological improvements (Kurokami et al. 2022) yielded a significantly higher accuracy rate of 70%.

successfully understood the presuppositional trigger (albeit not at fully adult-like levels). These results suggest that children's non-adultlike behavior in the Question-Answer Task should not be attributed to a lack of knowledge of *you*.

Future research might apply a similar paradigm to evaluate children's knowledge of other presuppositional triggers similar to *you* 'again'. More generally, we hope the present study offers methodological insights for investigating presuppositions and helps pave the way for further work exploring ways of felicitously tapping into children's knowledge of presupposition triggers.

Ethics statement

This study was approved by the ethics board at Tsinghua University (approval ID: THU2021031).

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